

Technical Document #155: Cloud Computing - Comprehensive Overview

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Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Auto-scaling automatically adjusts compute resources based on demand.
- Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet.
- Containerization packages applications with their dependencies.